

DeepSeek-V4 Architecture

Deep Dive into DeepSeek — Chapter 12

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The one rule of this chapter

V4 changes *what attention compresses* and *how the residual stream is mixed*.

It does **not** retire DeepSeekMoE, MTP, or the causal contract.

- Two operators carry the headline story: hybrid CSA/HCA attention over compressed KV state, and mHC's doubly-stochastic residual mixing.
- Everything else in the forward pass — DeepSeekMoE routing, multi-token prediction — is inherited from V3, unchanged by report.
- Training, agent behavior, and serving are **not** this chapter: they get their own chapters so optimizer state, policy behavior, and cache layout are never mistaken for checkpoint sublayers.

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Goal: follow one token through a V4 block — hybrid attention, mHC residual mixing, and the boundaries where operators must agree — and keep every number's evidence status visible.

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5. Summary and what carries forward

V4-Pro and V4-Flash in the family

Three coupled pressures, one preview list

- Long context must not inherit dense attention cost.
- Sparse capacity must not become dense per-token work.
- Deeper learned residual pathways must remain stable through depth.
- Specialized post-training must still yield one deployable policy.

The official report answers with three headline upgrades: hybrid [CSA/HCA](#) attention, [mHC](#) residual mixing, and the [Muon](#) optimizer [REPORTED: DeepSeek-V4 technical report].

- That is an upgrade hierarchy, not the entire model.
- V4 retains the Transformer, DeepSeekMoE, and MTP lineage — the report says the MTP configuration is unchanged from V3.

Pro and Flash: reported scale

Quantity	V4-Pro	V4-Flash
Total parameters	1.6T	284B

[REPORTED: DeepSeek API preview release note]

- The preview note names token-wise compression plus DSA as a central innovation for the whole family.
- These numbers establish scale and capacity. They do **not** determine workload quality or serving latency.

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Structural budgets, side by side

Field	V4-Flash	V4-Pro
Decoder stack	43 layers, hidden 4096	61 layers, hidden 7168

[INSPECTED: DeepSeek-V4-Pro / DeepSeek-V4-Flash config.json]

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CSA selector top- k	<code>index_topk=512</code>	<code>index_topk=1024</code>

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Both variants share the same mechanism family. They differ in depth, width, expert pool, and selector budget — not in kind.

Active parameter ratio

Definition

For total parameters P_{total} and per-token active parameters P_{active} ,

$$\rho = \frac{P_{\text{active}}}{P_{\text{total}}}.$$

Applying it to the reported counts: $\rho_{\text{Pro}} = 49/1600 \approx 3.1\%$ and $\rho_{\text{Flash}} = 13/284 \approx 4.6\%$

[DERIVED: reported total/active counts].

Pro's smaller ratio does not make it cheaper: it still activates 49B parameters per token, on a wider stack with a larger selector. Ratios describe conditional activation, not a product ranking.

Hybrid attention: CSA, HCA, and DSA

Three resolutions across the stack

SWA
exact recent window

preserves nearby dependencies; runs alongside
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[REPORTED: DeepSeek-V4 technical report].

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Compression changes representation. Selection changes which represented entries participate. Those are different knobs.

CSA and HCA, defined against each other

	CSA	HCA
Pooling	overlapping $2m$ -token span, stride m	non-overlapping m' -token groups
Rate	$m = 4$	$m' = 128$
After pooling	DSA top- k selection	dense attention over all entries

[REPORTED: DeepSeek-V4 technical report]

CSA trades a small, overlapping compression rate for learned sparsity. HCA trades a much larger compression rate for exhaustive, dense read. Both still cache shared-KV entries and run alongside SWA.

DSA, specialized to V4

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- V4 applies the same selection rule *inside CSA*, but *after* KV compression.

Its indexer now ranks *compressed* entries that summarize source spans, not the per-position latent entries selected in V3.2 [REPORTED: DeepSeek-V4 technical report; V3.2-Exp release note].

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That shift means selector recall and compression fidelity become two separate things to test — a selector can pick a legal index while the underlying compressed entry is a poor summary, or vice versa.

From residual connections to mHC

The rigid single lane, and the risk of relaxing it

Ordinary residual: $x_{l+1} = x_l + \mathcal{F}_l(x_l)$. The identity route is never multiplied by a learned map.

Hyper-Connections (HC) expand the stream to n lanes and learn pre-, post-, and residual maps around each sublayer — but an *unconstrained* residual map $\mathcal{H}_l^{\text{res}}$ composes across depth:

$$\mathcal{M}_{l \rightarrow L} = \mathcal{H}_{L-1}^{\text{res}} \cdots \mathcal{H}_l^{\text{res}}.$$

As a scalar illustration, a gain of only 1.05 repeated across 64 layers becomes $1.05^{64} \approx 22.7$

[DERIVED: scalar illustration, not a paper measurement].

The danger is a product, not one alarming layer. mHC is the fix.

mHC: the doubly stochastic constraint

Definition (doubly stochastic residual mixer)

A nonnegative matrix P with $\sum_j P_{i,j} = 1$ and $\sum_i P_{i,j} = 1$ for every row i and column j .

- mHC keeps HC's expanded residual lanes but constrains $\mathcal{H}_l^{\text{res}}$ to this manifold [REPORTED: DeepSeek-V4 technical report; mHC paper].

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This is residual-stream mixing. It is not a new attention mechanism and not a retrieval improvement.

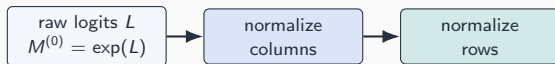
Reaching the constraint: Sinkhorn balancing

raw logits L
 $M^{(0)} = \exp(L)$

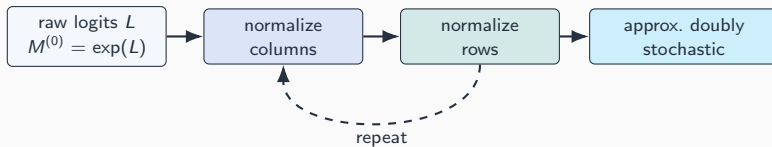
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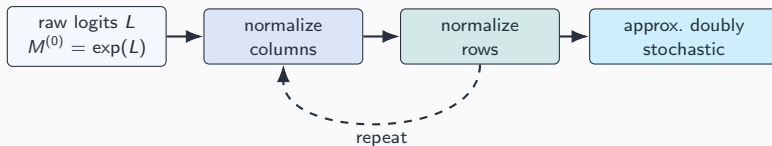
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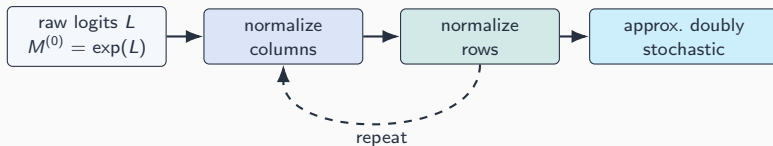


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Real kernels stop after `hc_sinkhorn_iters=20` alternations [INSPECTED: Pro/Flash config.json] — an approximate projection, not a symbolic guarantee that every sum is exactly one.

Composed across two sublayers: a numeric certificate

For an attention sublayer then a feed-forward sublayer, the direct residual route composes as

$$M = P_1 P_0.$$

- In a constructed four-lane, two-sublayer trace: $M\mathbf{1} = \mathbf{1}$, $\mathbf{1}^\top M = \mathbf{1}^\top$, and $\|M\|_1 = \|M\|_\infty = 1$ [DERIVED: worked architectural trace, not recovered V4 weights].
- A didactic unconstrained control $Q_l = 1.05 P_l$ already reaches $\|Q_1 Q_0\|_1 = 1.1025$ after only two sublayers.

At V4 scale, the official paper's 120-layer diagnostic reports unconstrained HC's composite gain approaching 3000, while mHC stays around 1.6, at a reported 6.7% marginal training overhead [REPORTED: mHC paper].

Those are training diagnostics, not a serving-latency or retrieval guarantee.

Architecture interactions and open questions

Where operators must agree

Boundary	Shared state that must agree
CSA/HCA + DSA	compressed entries, selector scores/indices, exact SWA state, positions
mHC + depth	four residual lanes, constrained mixing matrices
Attention + mHC	selected attention output, lane reads and writes
mHC + MoE	feed-forward input lanes, routed-token identity, residual write

A shape-correct implementation can still be wrong: legal tensors, wrong compressed span, selection, or lane boundary. [OPEN: joint architecture-interaction ablations]

Engram stays adjacent

- The released V4-Pro/Flash mechanism set is backed by the report and artifacts: mHC, CSA, HCA, DSA, DeepSeekMoE, MTP.
- Those sources mention Engram only as a [future direction](#), not a released component.

[REPORTED: DeepSeek-V4 technical report; Pro/Flash model cards; Engram repository/paper]

A source-family resemblance does not promote adjacent research into released architecture.

Summary and what carries forward

What this chapter established

- Hybrid CSA/HCA attention over compressed KV state, with an exact SWA path and a DSA selector specialized to compressed entries.

[REPORTED: architecture and evaluation]

[INSPECTED: Pro/Flash artifact and configuration surfaces]

[DERIVED: shape, compression, and residual traces]

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- Hybrid CSA/HCA attention over compressed KV state, with an exact SWA path and a DSA selector specialized to compressed entries.
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None of these structural facts alone ranks quality or serving speed — matched ablations remain open, not inferred.

Concept-to-code map for the chapter

- **Pro/Flash structural fields** → DeepSeek-V4-Pro/config.json, DeepSeek-V4-Flash/config.json
- **headline mechanism definitions** → DeepSeek-V4 technical report (DeepSeek_V4.pdf)
- **doubly stochastic residual mixer** → mHC paper, arXiv:2512.24880
- **released model class** → DeepseekV4ForCausalLM, Pro/Flash model cards

No DeepSeek-V4 inference repository is open-sourced at the time of writing — unlike V3, this chapter's evidence lives in the technical report and the pinned model-card configurations, not in inspectable kernel code.

Compress what attention reads. Constrain how the residual mixes.

Next: the training chapter builds these checkpoints — Muon, FP4 QAT, and specialist consolidation into one policy.